

Semiconductor packaging industry update

EQUITY: JAPAN ELECTRONIC PARTS

Packaging substrate shipments up 98% y-y

Rise in price per m² drives growth in shipments; tight supply–demand conditions for high-end package substrates unlikely to ease any time soon

Packaging-related monthly statistics: Record-high shipments in June

According to the *Current Survey of Production* for June, released on 17 August, shipments of packaging substrates (rigid module substrates) rose 98% y-y to ¥29.1bn, surpassing the previous record high of ¥27.8bn set in May. Shipments on a m² basis rose 20% and the average price per m² rose 65% to ¥1.429mn, beating the previous high of ¥1.356mn in May. It appears that shipments of packages for Rubin processors increased. Ibsiden [4062] (Buy) said that at the behest of customers, in Jul–Sep it will be prioritizing production of package substrates for server CPUs (ARM cores) over those for GPUs. We think the company plans to temporarily increase production of CPU package substrates, which are currently in short supply, because it has secured a certain amount of inventory of Rubin package substrates ahead of mass production of servers. It looks as though the company will resume increasing production of Rubin package substrates in Oct–Dec or later.

Easing of tight supply–demand conditions for high-end package substrates could take two to three years

Our impression from Apr–Jun results in the package substrate industry is that supply–demand conditions have been tightening for server CPUs, network switches, GPUs for AI data centers, and high-end package substrates used in custom ASICs. When supply–demand was tight in 2022, the world's largest CPU manufacturer, which had lost market share at the time, placed excessive orders for package substrates, for which supply–demand conditions were already tight, thereby suppressing CPU production volumes at rivals. This is one example of how substantial rush demand was generated due to customer circumstances, and it appears to have been unsustainable in some aspects. However, in the current case, production yields have fallen as a result of the shift to larger high-end package substrates, and with shipment volumes sluggish, demand volumes have been rising (sharp growth in demand for server CPUs and network switches as well as GPUs and AI accelerators), on which basis we think supply–demand conditions are likely to remain tight for the next two to three years. The packaging substrate industry has a limited number of customers, making it difficult to gauge the actual tightness of supply–demand conditions from outside, but given that only a few plants in the world are capable of producing advanced packages, we think conditions are already such that low-priority projects cannot be produced at plants that make cutting-edge products. These projects are manufactured at plants that produce older generation package substrates for top-tier manufacturers, mid-tier, and bottom-tier manufacturers, and therefore we think there could be production issues at the implementation stage going forward. If such problems were to come to the fore, it may be easier for people outside the industry to see the actual tightness in supply–demand conditions.

Looking ahead, we think a key point to watch in terms of packaging substrate supply capacity will be uptake of Embedded Multi-die Interconnect Bridge Through-Silicon Via (EMIB-T), which we expect to enter mass production in the second half of 2027. EMIB-T has longer production processes than regular high-end packages and is a highly difficult product that can only be made at advanced plants, on which basis we think mass production of these packages will reduce supply capacity at advanced plants. We think that Ibsiden, which has the largest number of advanced production lines in the industry, will be a major supplier of these packages, and we think that constraints on supply capacity will make it difficult for it to respond to major demands from customers in a timely fashion. We think the packaging substrate industry has entered a phase of competition in terms of supply capacity expansion as well as production technologies for large substrates.

Research Analysts

Japan electronic parts

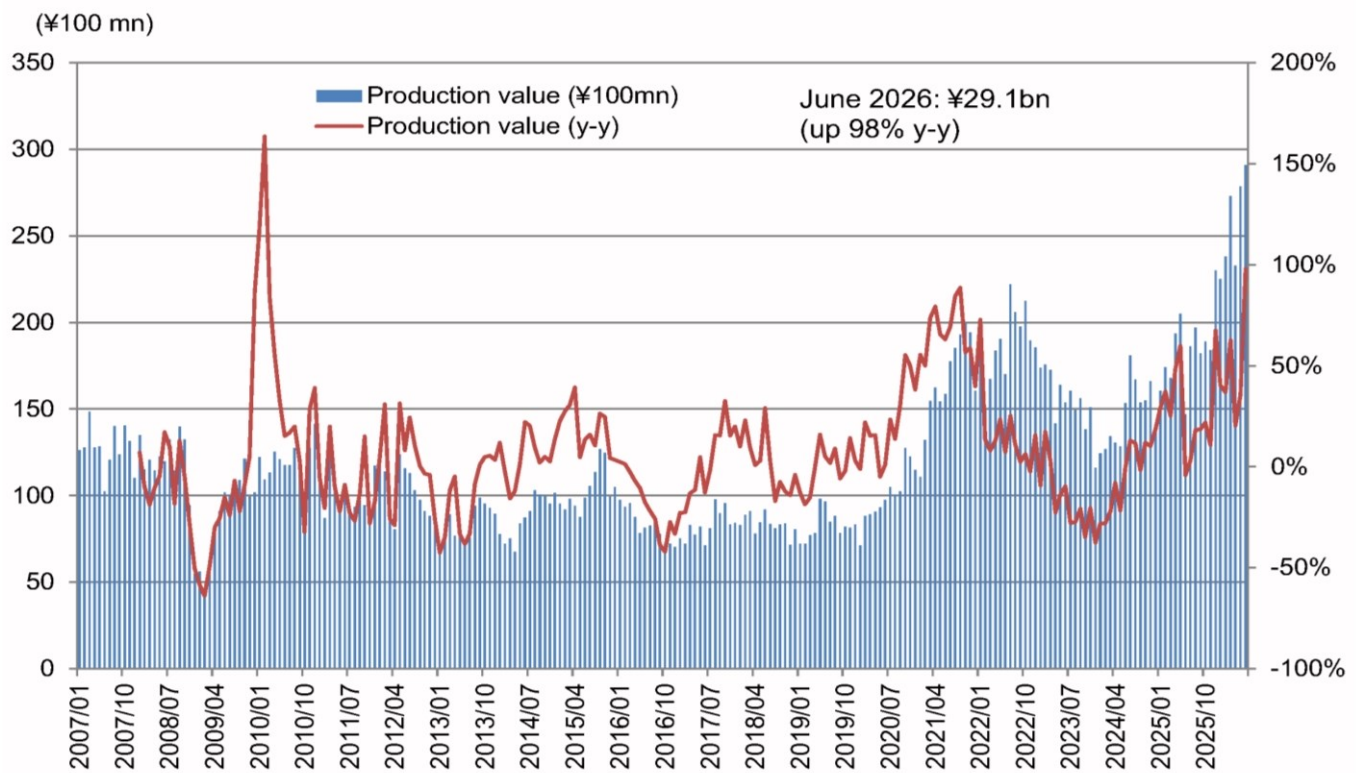
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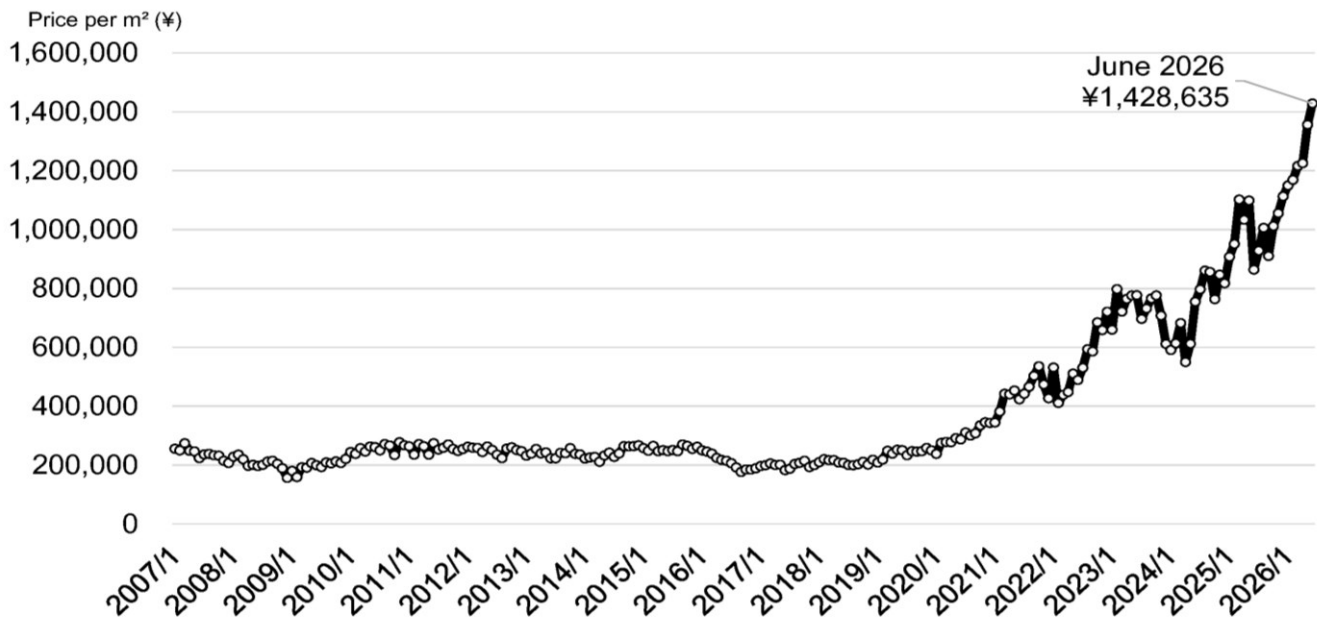
Reference figures

Fig. 1: Rigid PCB production value



Source: Nomura, based on Ministry of Economy, Trade & Industry's Current Survey of Production

Fig. 2: Price per m² for rigid module boards



Note: Module boards have components mounted on the board surface.

Source: Nomura, based on Ministry of Economy, Trade & Industry's Current Survey of Production

Appendix A-1

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Transactions involving Japanese equities (including Japanese REITs, Japanese ETFs, and Japanese ETNs, Japanese Infrastructure Funds) are subject to a sales commission of up to 1.43% (tax included) of the transaction amount (or a commission of ¥2,860 (tax included) for transactions of ¥200,000 or less). When Japanese equities are purchased via OTC transactions (including offerings), only the purchase price shall be paid, with no sales commission charged. However, Nomura Securities may charge a separate fee for OTC transactions, as agreed with the customer. Japanese equities carry the risk of losses owing to price fluctuations. Japanese REITs carry the risk of losses owing to fluctuations in price and/or earnings of underlying real estate. Japanese ETFs and ETNs carry the risk of losses owing to fluctuations in the underlying indexes or other benchmarks. Japanese Infrastructure Funds carry out the risk of losses owing to fluctuations in price and/or earnings of underlying infrastructures.

Transactions involving foreign equities are subject to a domestic sales commission of up to 1.045% (tax included) of the transaction amount (which equals the local transaction amount plus local fees and taxes in the case of a purchase or the local transaction amount minus local fees and taxes in the case of a sale) (for transaction amounts of ¥750,000 and below, maximum domestic sales commission is ¥7,810 (tax included)). Local fees and taxes in foreign financial instruments markets vary by country/territory. When foreign equities are purchased via OTC transactions (including offerings), only the purchase price shall be paid, with no sales commission charged. However, Nomura Securities may charge a separate fee for OTC transactions, as agreed with the customer. Foreign equities carry the risk of losses owing to factors such as price fluctuations and foreign exchange rate fluctuations.

Margin transactions are subject to a sales commission of up to 1.43% (tax included) of the transaction amount (or a commission of ¥2,860 (tax included) for transactions of ¥200,000 or less), as well as management fees and rights handling fees. In addition, long margin transactions are subject to interest on the purchase amount, while short margin transactions are subject to fees for the lending of the shares borrowed. A margin equal to at least 30% of the transaction amount (at least 33% for online transactions) and at least ¥300,000 is required. With margin transactions, an amount up to roughly 3.3x the margin (roughly 3x for online transactions) may be traded. Margin transactions therefore carry the risk of losses in excess of the margin owing to share price fluctuations. For details, please thoroughly read the written materials provided, such as listed securities documents or documents delivered before making a contract.

Transactions involving convertible bonds are subject to a sales commission of up to 1.10% (tax included) of the transaction amount (or a commission of ¥4,400 (tax included) if this would be less than ¥4,400). When convertible bonds are purchased via OTC transactions (including offerings), only the purchase price shall be paid, with no sales commission charged. However, Nomura Securities may charge a separate fee for OTC transactions, as agreed with the customer. Convertible bonds carry the risk of losses owing to factors such as interest rate fluctuations and price fluctuations in the underlying stock. In addition, convertible bonds denominated in foreign currencies also carry the risk of losses owing to factors such as foreign exchange rate fluctuations.

When bonds are purchased via public offerings, secondary distributions, or other OTC transactions with Nomura Securities, only the purchase price shall be paid, with no sales commission charged. Bonds carry the risk of losses, as prices fluctuate in line with changes in market interest rates. Bond prices may also fall below the invested principal as a result of such factors as changes in the management and financial circumstances of the issuer, or changes in third-party valuations of the bond in question. In addition, foreign currency-denominated bonds also carry the risk of losses owing to factors such as foreign exchange rate fluctuations.

When Japanese government bonds (JGBs) for individual investors are purchased via public offerings, only the purchase price shall be paid, with no sales commission charged. As a rule, JGBs for individual investors may not be sold in the first 12 months after issuance. When JGBs for individual investors are sold before maturity, an amount calculated via the following formula will be subtracted from the par value of the bond plus accrued interest: (1) for 10-year variable rate bonds, an amount equal to the two preceding coupon payments (before tax) x 0.79685 will be used, (2) for 5-year and 3-year fixed rate bonds, an amount equal to the two preceding coupon payments (before tax) x 0.79685 will be used.

When inflation-indexed JGBs are purchased via public offerings, secondary distributions (uridashi deals), or other OTC transactions with Nomura Securities, only the purchase price shall be paid, with no sales commission charged. Inflation-indexed JGBs carry the risk of losses, as prices fluctuate in line with changes in market interest rates and fluctuations in the nationwide consumer price index. The notional principal of inflation-indexed JGBs changes in line with the rate of change in nationwide CPI inflation from the time of its issuance. The amount of the coupon payment is calculated by multiplying the coupon rate by the notional principal at the time of payment. The maturity value is the amount of the notional principal when the issue becomes due. For J17 and subsequent issues, the maturity value shall not undercut the face amount. Purchases of investment trusts (and sales of some investment trusts) are subject to a purchase or sales fee of up to 5.5% (tax included) of the transaction amount. Also, a direct cost that may be incurred when selling investment trusts is a fee of up to 2.0% of the unit price at the time of redemption. Indirect costs that may be incurred during the course of holding investment trusts include, for domestic investment trusts, an asset

management fee (trust fee) of up to 5.5% (tax included/annualized basis) of the net assets in trust, as well as fees based on investment performance. Other indirect costs may also be incurred. For foreign investment trusts, indirect fees may be incurred during the course of holding such as investment company compensation.

Investment trusts invest mainly in securities such as Japanese and foreign equities and bonds, whose prices fluctuate. Investment trust unit prices fluctuate owing to price fluctuations in the underlying assets and to foreign exchange rate fluctuations. As such, investment trusts carry the risk of losses. Fees and risks vary by investment trust. Maximum applicable fees are subject to change; please thoroughly read the written materials provided, such as prospectuses or documents delivered before making a contract.

In interest rate swap transactions and USD/JPY basis swap transactions ("interest rate swap transactions, etc."), only the agreed transaction payments shall be made on the settlement dates. Some interest rate swap transactions, etc. may require pledging of margin collateral. In some of these cases, transaction payments may exceed the amount of collateral. There shall be no advance notification of required collateral value or collateral ratios as they vary depending on the transaction. Interest rate swap transactions, etc. carry the risk of losses owing to fluctuations in market prices in the interest rate, currency and other markets, as well as reference indices. Losses incurred as such may exceed the value of margin collateral, in which case margin calls may be triggered. In the event that both parties agree to enter a replacement (or termination) transaction, the interest rates received (paid) under the new arrangement may differ from those in the original arrangement, even if terms other than the interest rates are identical to those in the original transaction. Risks vary by transaction. Please thoroughly read the written materials provided, such as documents delivered before making a contract and disclosure statements.

In OTC transactions of credit default swaps (CDS), no sales commission will be charged. When entering into CDS transactions, the protection buyer will be required to pledge or entrust an agreed amount of margin collateral. In some of these cases, the transaction payments may exceed the amount of margin collateral. There shall be no advance notification of required collateral value or collateral ratios as they vary depending on the financial position of the protection buyer. CDS transactions carry the risk of losses owing to changes in the credit position of some or all of the referenced entities, and/or fluctuations of the interest rate market. The amount the protection buyer receives in the event that the CDS is triggered by a credit event may undercut the total amount of premiums that he/she has paid in the course of the transaction. Similarly, the amount the protection seller pays in the event of a credit event may exceed the total amount of premiums that he/she has received in the transaction. All other conditions being equal, the amount of premiums that the protection buyer pays and that received by the protection seller shall differ. In principle, CDS transactions will be limited to financial instruments business operators and qualified institutional investors. Transfers of equities to another securities company via the Japan Securities Depository Center are subject to a transfer fee of up to ¥11,000 (tax included) per issue transferred depending on volume. No account fee will be charged for marketable securities or monies deposited.

Nomura Securities Co., Ltd.

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